

Participatory Approaches to Sustainable Rural Development: A Study of Naala Village Uttarakhand

Krishnapriya V B¹, T J Navaneeth Krishnan¹, Aditya Bhatt B², Vivekananda Namani¹,
Amal Chandran J³, Vaishnav B Ajith²,
Sreevidhya C⁴, Vishnu Priya³, Renjith Mohan⁴

¹Department of Electronics and Communication Engineering, Amrita Vishwa Vidyapeetham,
Amritapuri, India

²Department of Electrical and Electronics Engineering, Amrita Vishwa Vidyapeetham,
Amritapuri, India

³Department of Mechanical Engineering, Amrita Vishwa Vidyapeetham, Amritapuri,
India

⁴School for Sustainable Futures, Amrita Vishwa Vidyapeetham, Amritapuri, India

Abstract

This paper presents a community-driven assessment of Naala Village using Participatory Rural Appraisal (PRA) techniques. The study was carried out through collaborative exercises including resource mapping, multiple transect walks, seasonal calendar development, and inflow–outflow analysis. These methods provided an in-depth understanding of how the community interacts with its environment, manages livelihoods, and addresses challenges related to water, sanitation, waste, and economic activities. The transect walks exposed recurring issues such as open drainage near living areas, poor waste disposal, and a shortage of clean drinking water, which directly impacts public health. Seasonal analysis revealed significant variations in agricultural productivity, disease occurrence, water availability, and livelihood stability throughout the year. The inflow–outflow framework highlighted key dependencies on external supplies for sanitation, healthcare, and energy, while also indicating self-sustained practices in farming and local services. Findings point to immediate needs for improving water infrastructure, enhancing waste management, and expanding livelihood options through polyhouse farming and skill development. The research not only captures the socio-economic fabric of the village but also outlines actionable strategies tailored to community needs, offering a participatory model for rural sustainability and resilience.

Keywords: Participatory Rural Appraisal (PRA), Human-Centered Design (HCD), Sustainable Rural Development, Water Infrastructure, Community Engagement

1. Introduction

Rural communities often face interconnected challenges related to water scarcity, waste management, unstable livelihoods, and limited infrastructure. Addressing these issues requires a thorough understanding of local resource dynamics, seasonal patterns, and socio-economic conditions. Participatory Rural Appraisal (PRA) is a widely adopted approach that enables communities to collaboratively map their resources, identify problems, and propose locally viable solutions.

This study focuses on Naala Village in Uttarakhand, where participatory tools such as transect walks, resource mapping, seasonal calendars, and inflow-outflow analysis were employed. Conducted as part of the Amrita Live-in-Labs[®] program, the study involved immersive field engagement aimed at capturing the community's perspective on critical issues such as water management, sanitation gaps, waste handling, and livelihood sustainability. Unlike conventional surveys, this participatory approach ensures that the solutions are deeply rooted in the community's needs, capacities, and experiences. The findings aim to support targeted interventions that improve living conditions while promoting sustainable rural development.

2. Study Area

Naala is a scenic village in Rudraprayag district of Uttarakhand state on the Kedarnath road, 17 km from Ukhimath and 44 km from Rudraprayag town. The village area is 181.5 hectares and it is renowned for its terracing rice and wheat fields, which denotes its agriculturally based habitat.

Naala has 1,189 people who reside in 257 families as per the 2011 Census, a sex ratio of 982 females for every 1,000 males over the state average. The literacy rate is 83.89%, though there is huge gender gap: male literacy at 96.6% and female literacy at 71.05%. Livestock rearing and agriculture are the principal sources of livelihood, farming and animal rearing being centrally run by women. Large groups of men earn seasonal jobs for six months of the Kedarnath Yatra season, employed as drivers, porters, shopkeepers, or tent workers earns income in their homes. The village is religiously oriented, which revolves around the 900-year-old temple of Maa Tripura Sundari. Based on its peaceful landscape, Himalayan landscape, and seasonal cycle marked by the pilgrimage season, Naala presents a balance between cultural heritage, natural landscape, and socio-economic reliance on seasonal work. Gender imbalance, shortage of infrastructure, and seasonal exposure, however, are the major issues.



Figure 1: Naala Village, Uttarakhand

3. Methodology

Participatory Rural Appraisal (PRA) is a set of participatory, visual, and field-based methods that are applied in engaging rural communities in analyzing their socio-economic and environmental reality. It helps local populations to state their needs, identify priorities, and unanimously come up with context-specific solutions using indigenous knowledge and experiences.

Different PRA tools like resource mapping, seasonal calendars, transect walks, and problem trees are deployed for collaboratively assessing spatial, temporal, and structural dynamics. When related to sustainability, PRA works at the community level on various interconnected themes such as water, agriculture, health, and livelihoods. The key strength of PRA lies in highlighting the narrative of a community at the macro level, so that researchers and practitioners can comprehend broader patterns, systemic dependencies, and common priorities. In the process of problem identification and solution framing, PRA emphasizes the use of visual tools and group activities to assure inclusivity so that even the marginalized voices find their place

Resource mapping as shown in Fig. 2 was conducted through a participatory approach combining field observations, informal interviews, and community sketching. A participatory sketching session was then held with two local villagers and the village coordinator to collaboratively map these features. Key village assets such as water tanks, schools, temples, and agricultural zones were identified and mapped with the help of residents and the local coordinator. GPS tools were used to validate spatial accuracy. The map provided a visual understanding of the village's resource distribution and infrastructure layout.

The transect walk as shown in Fig. 3 was used with the aim of comprehending how community members may be utilizing the resources and spaces in the village. Transect walks enables to observe directly the day-to-day practices and spatial patterns across different zones of a village. These walks were guided by the residents and the village coordinator who provided contextual insights. The observations were of service infrastructure in use, environmental conditions, resource accesses, and these observations were duly noted. To analyze the seasonality of agriculture, water availability, disease occurrence, and livelihood fluctuations in Naala village, a seasonal calendar Fig. 4 was developed using participatory observation and focus group discussions. Initially, the sketch was prepared with villagers from the village, after which it was supplemented with expert inputs from the Agricultural Officer and the PHC Staff. Following data triangulation, the information was systematically categorized into key sectors such as Agriculture, Energy, Livelihood, Water, Health, Education, Waste, Natural Calamities, and Season. Each activity or issue was mapped on a 12-month matrix using a color-coded system (Red = High, Yellow = Moderate, Green = Low, White = Negligible) to indicate seasonal intensity or occurrence.

To analyze resource dependencies and financial flows in Naala village, the PRA tools Inflow - Outflow and Income - Expenditure Fig. 5 were used through a participatory approach. Drafts were collaboratively developed with experienced villagers through Focus Group discussions and improved by engaging stakeholders like the village head, health staff, and Anganwadi worker through consultations. Group discussions and household surveys were used to triangulate data, and visual mapping techniques were used to ensure inclusive participation. The matrices developed gave insights into internal capacities, external dependencies, and economic patterns at the household level.

A participatory stakeholder mapping Fig. 6 exercise was conducted to visualize the relative proximity and influence of key institutions around Naala village. Semi-structured interviews and household surveys helped identify essential services such as government offices, welfare centers, and agricultural hubs and assessed usage frequency and perceived importance. Institutions were placed on the map based on access and relevance, with circle sizes reflecting community dependence. The map was validated and refined along with the villagers.

In order to synthesize insights obtained from the different PRA tools like resource mapping, seasonal calendars, and inflow-outflow analysis, a systematic brainstorming exercise was held among community stakeholders. These consisted of village women, the Anganwadi worker, and local health staff. The results from the previous tools and exercises were presented for a free discussion to confirm problems witnessed and reveal underlying causal linkages.

The Problem Tree (PT) as illustrated in Fig. 7 was jointly created to define the root causes, key problems, and impacts associated with the unavailability of clean drinking water. Causes like inadequate pipeline maintenance, absence of filtration, and spatial unavailability of natural water resources were mapped systematically. Through this process of participation, not only infrastructural and environmental causes but also behavioral and economic impacts like added disease burden and lost time for fetching water were identified. The draft PT was then sharpened through subsequent interactions and tested with the same stakeholders to ensure if it reflected local realities. The validation process was critical to correctly reflect the interrelated issues encountered by marginalized homes, specifically those based on unsafe piped water in light of geographical exclusion from natural spring benefits. Personas and scenarios were created by combining ethnographic fieldwork with participatory tools. Semi-structured household interviews, outcomes of problem tree, focus group discussions, and direct observation were used to realize daily routines, water access behavior, and emotional responses associated with consuming water. Insights were synthesized into fact based personas representing typical user types, which were later validated through community feedback. Scenarios were built as stories demonstrating how various personas coped with water-related challenges. They were developed from group discussions, situated through field observation and interview findings. Each scenario captures goals, obstacles, and coping mechanisms of users and provides a peoplecentered point of view for designing participatory interventions

4. Results

This section outlines the key findings of the study and shows how each tool within the Participatory rural appraisal and Human centric design contributed to a deeper understanding of the community dynamics and their core challenges.

4.1. Resource Mapping of Naala Village

The resource map of Naala village presents a detailed spatial representation of the community's infrastructure, land use patterns, and resource distribution. The village is structured around a natural spring (Dhaara), which plays a crucial role in meeting domestic water needs. Residential settlements are densely clustered near the main road and central junctions, where essential facilities such as the temple, school, Anganwadi

centre, government food processing unit, playground, retail shops, and petrol pump are located. This concentration indicates the socio-economic core of the village.

The village's topography significantly influences its spatial arrangement and resource accessibility. Geographically, the western region is situated at a relatively higher elevation, gradually sloping down toward the eastern side. The terrain exhibits a stepped pattern typical of mountain foothill settlements, with terraced formations used extensively for agriculture. This step-like terrain supports the cultivation of rice, wheat, and vegetables, with rice fields predominantly located in lower lying areas closer to water sources, while wheat and vegetable farming extends across mid-elevation and higher terraces.

Agricultural activity is predominantly concentrated in the southern, western, and southwestern regions, where fertile lands support diverse cropping systems. The presence of polyhouse farms, especially in the western highland zones, suggests the community's partial adoption of controlled environment agriculture to improve yield stability against climate variability.

Water infrastructure displays a notable spatial pattern. A primary water tank is positioned in the southwestern part, serving nearby residential clusters. In contrast, households in the southeastern and northeastern sectors rely heavily on natural water collection point, leading to longer travel distances for water access.

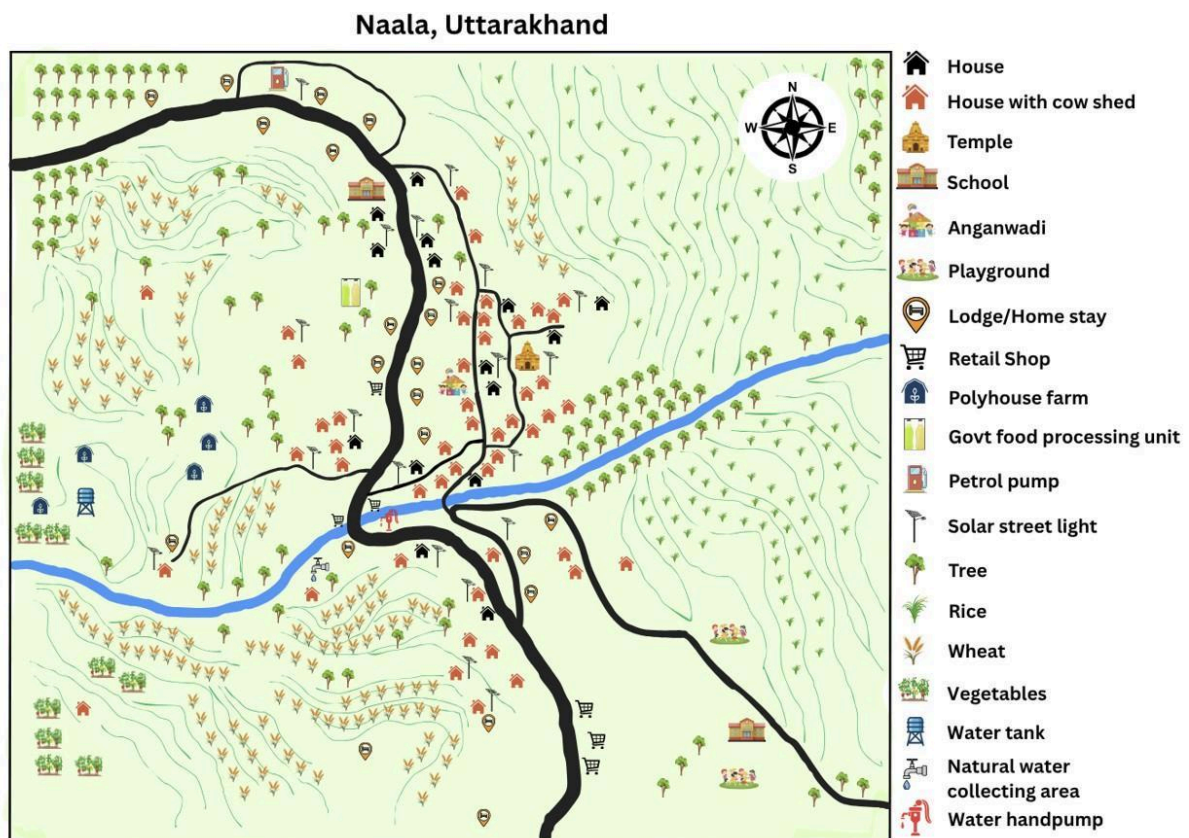


Figure 2: Resource Map of Naala Village, Uttarakhand

4.2. Transect Walk

Three transect walks were conducted across different parts of Naala village. Among these, Transect Walk 1 proved to be the most significant, as it covered the major residential and communal zones of the village and played a crucial role in identifying key infrastructural and environmental challenges. During our transect walk in Naala village beginning at the tuition center and ending at a house, we explored various landmarks including farmlands, local shops, community households, and a small hotel. Each location revealed its own strengths and shortcomings the tuition center functioned as an educational institute agricultural fields showed signs of productivity, but crops are damaged by stray animals shops and hotels offered essential services, but exhibited gaps in hygiene and waste disposal. The concern developed in residential areas, where the lack of clean drinking water was evident. This was largely attributed to unsanitary waste management, open drainage systems near water sources, and a lack of basic purification methods. Addressing this problem requires practical interventions such as community based filtration systems, improved waste segregation, and awareness drives of safe water practices to improve the overall well-being of the village.

	 TUITION CENTER	 FIELD	 SHOP	 HOUSE	 HOUSE	 HOTEL	 HOUSE
LIVELIHOOD	Teach for income	Grain income	Retail jobs	Animal husbandry	Animal husbandry	Tourism jobs	Animal husbandry
EDUCATION	Skill development	Basic farming skills	Shop support	Basic education	Basic education	Tourism skills	Basic education
AGRICULTURE		Seasonal farming	Shop farming	Animal care	Animal care	Tourism, animals	Animal care
ENERGY & WASTE MANAGEMENT	Clean energy center	No packaging waste	Shop waste issues	Poor waste	Poor waste	Hotel waste issues	Poor waste
WATER & SANITATION	Clean center	Irrigation varies	Shop hygiene varies	Poor sanitation	Poor sanitation	High water use	Poor sanitation
PROBLEMS	Few jobs, unemployment	Crop damage	Seasonal sales issue	Poor waste, health	Poor waste, health	Low season income	Poor waste, health
OPPURTUNITIES	More education	Food security	Market growth	Healthier waste	Healthier waste	Expand, events	Healthier waste

Figure 3: Transect Walk 1

4.3. Seasonal Calendar and Inflow-Outflow Analysis

The seasonal calendar analysis of Naala village revealed key patterns in agricultural activities, resource usage, public health, and community mobility, offering vital insights into timesensitive vulnerabilities and opportunities. With the help of knowledgeable villagers, we identified the following:

In terms of energy, firewood and biomass usage peak during winter months (December–February), indicating reliance on traditional room heating sources. Power outages lasting over three hours were more frequent in summer, suggesting infrastructure strain due to elevated cooling demands. Livelihood patterns showed a heavy surge in out-migration

between March and July, coinciding with reduced agricultural activity and limited local employment. Seasonal tourism, particularly during the Kedarnath Yatra (April to November), contributes short-term livelihood opportunities. Water-related challenges were prominent during summer, with a noticeable increase in scarcity of clean drinking water. This led to greater reliance on external sources like water tankers, piped water, and packaged water. Health data revealed seasonal spikes in diseases; monsoon months saw a rise in waterborne illnesses such as typhoid and diarrhea, correlating with poor sanitation and contaminated water. Natural hazards like landslides and road blockages mainly occurred during the rainy season, disrupting transportation, healthcare access, and supply chains. Waste management and educational access remained relatively stable but were indirectly affected by migration and infrastructure breakdowns.

Following triangulation, data were categorized under sectors such as Agriculture, Energy, Livelihood, Water, Health, Education, Waste, Natural Calamities, and Season. Each activity or issue was mapped on a 12-month matrix using a color-coded system (Red = High, Yellow = Moderate, Green = Low, White = Negligible) to represent seasonal intensity. The calendar Fig. 4 was validated with villagers and adjusted based on their feedback and inputs from local officials.



Figure 4: Seasonal Calendar of Naala Village, Uttarakhand

The inflow-outflow analysis highlighted the community's balance between internal capabilities and external dependencies in sectors such as agriculture, health, sanitation, energy, and livelihood.

During peak tourism and summer months, water scarcity led to increased dependency on tanker and piped supplies. Though sanitation remains weak, community-led construction (e.g., through women's groups and rural builder initiatives) indicates promising progress. In agriculture, seeds, tools, and fertilizers are mostly imported, while local produce and niche activities like fruit squash and silkworm cultivation suggest emerging self-reliance. Healthcare shows partial dependency, especially for supplies and skilled staff. Education expenses, especially for higher studies, contribute to external financial outflows. Energy sources are mixed: LPG, kerosene, petrol, and electricity are externally sourced, while biomass and firewood

remain prominent. Waste management lacks structured systems, highlighting a critical gap. Seasonal income from tourism (e.g., homestays, shops, transport) contributes the inflow. All findings were validated with stakeholders such as the ward in-charge and village elders.

4.4. Income-Expenditure and Venn-diagram

Naala village income and expenditure patterns show a seasonal dependence and financial risk pattern. There are some stable income sources, but a major part of the livelihood is dependent on the Kedarnath pilgrimage season, which offers seasonal jobs like transport services, homestays, and casual labor. Outside peak times, income sources like agriculture and dairy are the main sources of income and mostly earned by women. Despite this, basic expenses on food, fuel, education, telecommunications, and health continue unabated, exerting constant pressures on family budgets.

The Venn diagram Fig. 6 mapped five different clusters of institutions near Naala village, each with different access and interaction patterns. At the core of everyday life are government and welfare institutions like the Pradhan's office, Anganwadi center, Self Help Group and ration shop, all located within the boundaries of the village. Their placement at the center and large sizes emphasize their role as readily accessible locations for daily administrative, child care, and food security needs. Immediately beyond the

Table 1: Inflow-Outflow Analysis of Naala Village

Category	Inflow	Outflow
Water	Water supply during Yatra season (tankers, piped water)	Nil
Sanitation	Sanitary products, cleaning agents, infrastructure (e.g., toilets, sewage pipes)	Nil
Health	Medical supplies, visiting doctors	Nil
Hygiene	Hygiene kits (soap, shampoo, toothpaste, pads)	Nil
Agriculture	Seeds, tractors, fertilizers, pesticides	Small-scale crop and fruit squash output
Risk Management	Road clearance during landslides	Nil
Waste Management	Nil	Nil
Education	Textbooks, supplies, college-related costs	Nil
Livelihood	Income from homestays and tourism	Labor (horse riding, hotels, transport)
Skill	Nil	Tailoring by SHG members
Energy	LPG, kerosene, petrol, electricity	Nil
Environment	Nil	Nil

Others	Rations, poultry, festival spending	Pickling, silkworm cultivation
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village boundaries are health and educational facilities, such as the Primary Health Centre, a village private clinic, and two schools - one government and one private. Outside the village's central area, there are the bank and post office as the central financial and communication hubs; their proximity encourages frequent use but still requires an intentional trip outside of residential groups. Outside the surrounding vicinity, support services for agriculture and economies such as the Agriculture Office, Horticulture Extension, food processing facility, and petrol pump—are in surrounding towns. Placed on the periphery, their presence reflects a dependence on outside resources for agricultural inputs, technical advice, and fuel.

4.5. Brainstorming

The brainstorming sessions, or participatory semi-structured focus group discussions, aimed at validating the PRA tools' findings and probing more deeply into community perceptions of water issues. Every session was pre-meditated with a series of open-ended guiding questions that permitted free sharing of ideas while ensuring discussions on important topics like access to water, filtration practices, and social inclusion. Participants were women from different caste groups, including the Anganwadi teacher and staff from Guptkashi PHC, who were selected to represent different viewpoints and potential levels of acceptance of the filtration system within the community. A trusted local coordinator was present throughout the sessions to create a comfortable atmosphere for participants while reducing language and cultural barriers. The brainstorming with the community, consisting of eight village women and the local coordinator, was conducted to validate primary findings from the resource map, seasonal calendar, and inflow-outflow chart. The discussion further reaffirmed earlier observations that included issues of water access, dependence on firewood supply, and seasonal unemployment that occurred

INCOME/EXPENDITURE (NAALA)		
Category	Income	Expenditure
Water		Rs 500/ 3 month
Gas(LPG)		Rs.1000/cylinder
Employment	Rs 10000/ month	
Electricity		Rs 900/month
Farming		Rs 600/ season
Dairy Products	Rs 6000/month	
Livestock and animal care	Rs 8000/ Day	Rs 700/ Purchase
Grocery		Rs 5000/month
Mobile bills		Rs 900/ month
Healthcare		Rs 500/ month
Education		Rs 1000/month
Squash	Rs.150/1L bottle	

Figure 5: Income-Expenditure details

post the Kedarnath Yatra. The villagers' comments led directly to the further development of the problem tree and the realisation that the focus should be on clean water. And they talked about what sort of water filter would be most appropriate for their village, whether for individual homes, for clusters of homes, or for a community as a whole. The session also showed community interest in biogas, if support and training are provided. A brainstorming meeting was held at Guptakashi PHC Fig. 8 with the Medical Officer and nursing staff, and it was found that there was high patient inflow (12,000–15,000 patients per annum), including seasonal illnesses such as typhoid and respiratory infections due to poor nutrition and unhealthy drinking water. Staff members raised concerns about equipment shortages and limited staff capacity. Similarly, a session with the Anganwadi teacher highlighted challenges in maternal and child health. The center monitors 30 children under six using the Poshan Tracker app but operates with poor infrastructure, including no electricity, access to proper toilets and handwashing areas, and roof leakage. The teacher noted that lack of clean drinking water contributes to frequent illnesses among children. Overall, the participatory brainstorming sessions helped validate field data and refine the Problem Tree, particularly on lack of clean drinking water. They highlighted health impacts such as illness, disease burden and the importance of better water treatment, sanitation, and awareness to address the community's clean water challenges. Participants collectively emphasized the need for a reliable, community-based filtration solution that is socially inclusive and easy to maintain and support for ongoing maintenance to ensure long-term sustainability and impact.

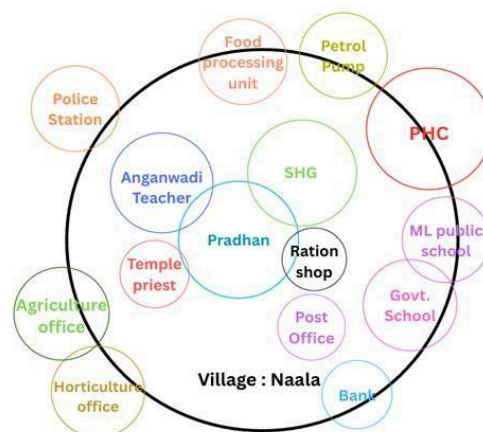


Figure 6: Venn diagram - Stakeholder mappings

4.6. Problem Tree Analysis

The lack of access to clean drinking water in Naala village stems from a combination of inadequate water infrastructure, absence of treatment facilities, and geographical barriers. The problem tree Fig 7 highlights critical root causes, their consequences, and the broader systemic impact on the community's health and well-being.

At the foundational level, the absence of proper maintenance of existing water infrastructure, including pipelines and storage tanks, emerges as a primary contributor. Field observations and interviews with village residents confirm that many pipelines are

aged, prone to leakages, and often clogged with silt and biological contaminants. Furthermore, the only community water tank lacks a functional treatment system. This tank typically stores water sourced from mountain springs, but does not incorporate even basic filtration or disinfection processes, making the water highly susceptible to microbial contamination, particularly during the monsoon and summer seasons when water scarcity is at its peak.

A significant geographic factor that exacerbates the problem is the pattern of settlement of the village. Many households are located in distant parts of the village, particularly in the lower regions, requiring residents, especially women and the elderly, to undertake long and physically taxing walks to fetch water from the natural spring. This challenge is not only logistical, but also has socio-economic repercussions. The time and labor expended on water collection reduce opportunities for income-generating activities and education, particularly affecting women and children.

Compounding the infrastructure deficits is the limited awareness of waterborne diseases and the risks associated with consuming untreated water. Despite periodic health campaigns, household interviews reveal that most residents do not practice water boiling or filtration consistently. This gap in health literacy correlates with the observed increase in diseases such as typhoid, diarrhea, and skin infections, especially during the summer, when the water sources are stressed and the risk of contamination peaks.

One direct consequence is the noticeable foul smell in piped water which is frequently reported during household surveys. These physical indicators serve as both a symptom and a warning of the microbial contaminant present. Consequently, a marginal portion of the population resorts to purchasing bottled water from nearby stores despite the high financial burden this imposes on low-income families. This reliance on external water sources indicates not only a health problem but also an economic strain that disproportionately impacts vulnerable groups.

Medical data obtained from the local Primary Health Centre confirms the link between unsafe water and public health outcomes. The PHC reports that approximately 40% of outpatient cases during peak seasons are attributed to waterborne diseases. Moreover, seasonal trends show a spike in these illnesses during the pre-monsoon and summer months, correlating with reduced spring water availability and greater dependence on compromised water storage systems.

These factors collectively create a cycle of reliance on untreated water sources and unreliable supply systems, contributing to persistent health risks and economic instability. Focus group discussions with women revealed constant stress over water availability during dry months. Systemic issues in policy and governance further worsen the problem. While government schemes exist for water infrastructure development, their reach is inconsistent. Field data indicates that bureaucratic delays, lack of technical support, and inadequate community engagement result in poorly implemented and incomplete projects. Additionally, villagers report minimal involvement in decision-making processes regarding water resource management, which undermines the sustainability of externally driven interventions.

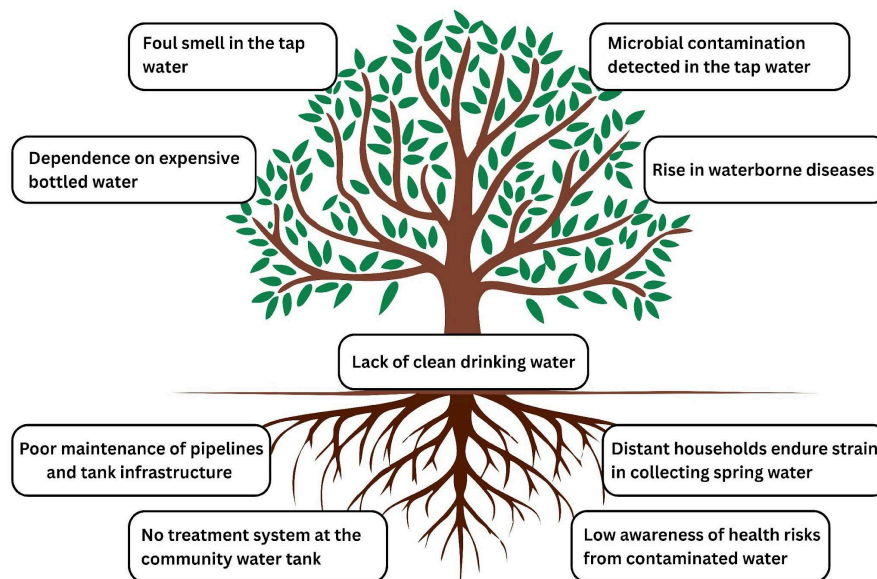


Figure 7: Problem Tree Analysis of Naala Village

4.7. Personas and Scenarios

To contextualize the challenges concerning water supply in Naala, user personas were developed based on field interviews with affected households. These reflect the lived experiences, routines, and pain points of those most impacted by water inaccessibility. The following table presents one such persona, followed by a scenario illustrating the implications in daily life.

Table 2: User Persona: Kamla Devi – Water Collection Burden

Name	Kamla Devi
Gender	Female
Description	60-year-old grandmother living in the lower region of Naala village with her son's family. She is responsible for managing household chores and looking after her three school-going grandchildren.
Tasks in Her Life	- Fetches water from the uphill <i>dhara</i> 3–4 times daily - Prepares meals, washes clothes, cares for grandchildren - Occasionally works as a daily wage labourer during the farming season

Pain Points	• Piped water at home is foul-smelling and visibly dirty, so she avoids using it for drinking • Long uphill walks to collect water cause physical fatigue and backpain • Grandchildren also assist with water collection, impacting their school time and energy • Daily trips reduce time for rest, paid work, or family care
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Kamla Devi begins her day before sunrise, heading uphill with buckets to the dhara. The piped water at home smells foul and contains dirt, so she makes 2 to 3 trips daily. Despite her backache, she has no choice. Her grandchildren often miss study hours as they help with water collection. This daily burden leaves her exhausted and eats into time that could be used for income generation or rest. The family's reliance on untreated and distant sources highlights the urgent need for improved, accessible, and clean water infrastructure.

5. Discussion

This research of Naala village provides an in-depth interpretation of rural water issues that differ from many generally reported problems in other Indian villages. The application of PRA and HCD tools served to disaggregate these dynamics. PRA techniques such as transect walks and seasonal calendars allowed a shared community vision of patterns of resource utilization and health risk, whereas HCD interviews emphasized the burden on individuals especially children and women, who are coping with water collection and sickness daily. But there were also practical limitations. PRA consensus-building was sometimes enabling decision making because various groups in the community identified different priorities. These results indicate the necessity for more adaptive and integrated models that enable participatory approaches to progress towards faster action.

The participatory assessment of Naala village revealed that the core water -related challenge is not the lack of infrastructure, but rather its poor maintenance and underutilization. Unlike many other rural communities in India that struggle with the absence of basic water access, Naala is comparatively well equipped with a community water tank and piped connections extending to almost all households. However, biological contamination detected through field water quality tests is prevalent due to inadequate upkeep of water storage tanks, pipe leakages, and lack of treatment facilities. A key inference from the problem tree analysis is that although the source water is chemically and physically clean, contamination occurs en route, particularly during storage and distribution. This results in a situation where infrastructure exists but without safe water access. Moreover, many villagers do not practice safe water handling procedures such as boiling or filtering leading to the consumption of untreated water and recurring occurrences of typhoid, diarrhea, and other waterborne illnesses. This is compounded by low health literacy and lack of community awareness around waterborne diseases.

One unique insight from the research in Naala village is the high level of cultural and emotional attachment that villagers have with their water sources. In contrast to most other rural villages where suspicion or fear for water quality is more prevalent, individuals in Naala show a strong faith in the natural purity of water, especially that derived from mountain springs. This religious and cultural reverence is a challenge to awareness of biological contamination as community members resist the notion that something they hold to be pure could be harmful. To address these challenges, a combination of technical upgrades, behavioral change, and sustainable infrastructure is proposed. These solutions are designed to be community-managed, context specific, and aligns with the government programs such as the Jal Jeevan Mission.



Figure 8: Field visit with PHC Medical Officer and Chief Nurse at Naala

Addressing the water-related challenges in Naala village requires a set of interventions that are both contextually grounded and community inclusive. While the village has a well-established piped water network and a central storage tank, the lack of water treatment and maintenance has led to widespread biological contamination, making the water unsafe for consumption. Therefore, the primary solution proposed is the installation of a community level filtration system. This centralized system has to incorporate ultraviolet (UV) filtration to remove microbial pathogens, as well as a pre-sediment filter to remove suspended particulates like silt and trash. A common filtration plant not only provides safe drinking water for everyone, but also serves to bridge the disparity that families using untreated sources presently endure. Besides filtration, routine maintenance of the water infrastructure has to be formalized. The field observations suggest that the majority of the current problems like bad smell and observable impurities stem from leaking and poorly maintained pipelines and tanks. There needs to be a regular maintenance schedule, at the behest of the village panchayat or a specialized water committee, to ensure long-term system dependability. This committee can be trained by local government authorities or development agencies to undertake basic repairs, water quality monitoring, and to take preventive measures before issues arise. As a complementary measure, rainwater harvesting systems can be encouraged at the domestic level, especially for dwellings located at farther distances from the community water supply source. Rooftop harvesting during the monsoon period can serve as an additional source of drinking water. Yet, this must be combined with straightforward point of use filtration systems so that the water harvested is safe to

drink. Although the household-level systems can provide more individual control, they are relatively more costly to set up and operate and can thus be rolled out progressively or targeted towards the most water insecure households

6. Conclusion

This study provides a detailed understanding of Naala Village's resource dynamics, livelihood patterns, and key challenges through a participatory approach. The use of tools like resource mapping, transect walks, seasonal calendars, and inflow-outflow analysis revealed pressing issues such as poor water access, inadequate waste management, and seasonal livelihood gaps. The findings highlight how infrastructural limitations, geographical barriers, and governance failures contribute to recurring health risks and economic strain.

Despite these challenges, the community shows potential for sustainable growth through targeted interventions like improved water infrastructure, better waste handling, expanded polyhouse farming, and skill-based employment. The participatory process not only uncovered ground realities but also fostered community-driven solutions, offering a practical framework for future development efforts.

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References

- [1] Krishnendu, S., Rohra, H. A., Kanakasabapathy, P., Nandan, K., and Padil, V. V., "Towards grassroots sustainable development using human centered design and participatory rural appraisal: A study in two rural Indian villages," *Sustainable Futures*, vol. 9, p. 100604, 2025.
- [2] Al-Taai, S., "Sustainable development—A study of its goals, dimensions and methods of achieving," in *IOP Conf. Ser.: Earth Environ. Sci.*, vol. 722, no. 1, p. 012028, IOP Publishing, 2021.

- [3] Narayanasamy, N., *Participatory Rural Appraisal: Principles, Methods and Application*, SAGE Publications Ltd., 2009.
- [4] Talgorn, E., and Ullerup, H., “Invoking ‘empathy for the planet’ through participatory ecological storytelling: From human-centered to planet-centered design,” *Sustainability*, vol. 15, no. 10, p. 7794, 2023.
- [5] Ramesh, M. V., Mohan, R., and Menon, S., “Live-in-Labs: Rapid translational research and implementation-based program for rural development in India,” in *Proc. IEEE Global Humanitarian Technology Conf. (GHTC)*, pp. 164–171, IEEE, 2016.
- [6] Varma, D. S., Nandan, K., PC, V. R., Perez, M. L., KA, S., and Ramesh, M. V., “Participatory design approach to address water crisis in the village of Karkatta, Jharkhand, India,” *Technol. Forecast. Soc. Change*, vol. 172, p. 121002, 2021.
- [7] Ajith, V., Reshma, A. S., Mohan, R., and Ramesh, M. V., “Empowering communities in addressing drinking water challenges using a systematic, participatory and adaptive approach and sustainable PPP model,” *Technol. Forecast. Soc. Change*, vol. 185, p. 121970, 2022.
- [8] Makadi, Y. C., Gadde, A., Patel, A., Adusumilli, T., Paladugu, S., Sankranti, S., and Cornet, S., “Building resilient rural communities: Participatory approaches in Khattukhal Village for sustainable development,” in *Proc. 2023 Int. Conf. Recent Adv. Info. Tech. Sustainable Dev. (ICRAIS)*, pp. 1–7, IEEE, 2023.
- [9] Basu, M., DasGupta, R., Hashimoto, S., and Hoshino, S., “A multi-actor and bottom-up perspective on attaining rural water security: qualitative evidence from India,” *Environ. Dev. Sustain.*, vol. 23, pp. 1461–1484, 2021.
- [10] Malan, A., and Sharma, H. R., “Assessment of drinking water quality and various household water treatment practices in rural areas of Northern India,” *Arab. J. Geosci.*, vol. 16, no. 1, p. 96, 2023.
- [11] Anoop, S., Adwaith, A., Aswin, A., Raja, S. K., Tejasvin, V., Nayan, M. K., & Nithin, P. M. (2024, December). Assessing the Participation of Villagers in a Participatory Rural Appraisal Based Study: An Amrita Live-In-Labs® Project Case Study. In *2024 IEEE 4th International Conference on ICT in Business Industry & Government (ICTBIG)* (pp. 1-5). IEEE.
- [12] Raman, R., Nedungadi, P., Amrita, R. P., Banerji, S., Mohan, R., and Ramesh, M. V., “Investigating the factors affecting the adoption of experiential learning programs: MBA students experience with Live-in-Labs,” in *Proc. 2020 IEEE Bangalore Humanitarian Tech. Conf. (B-HTC)*, pp. 1–6, IEEE, 2020.